

Smart Irrigation Prediction — CRISP-DM Final Report

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1. Dataset Characteristics

- **Environmental sensor records:** 10,964
- **Soil sensor records:** 32,666
- **Water meter records:** 32,647
- **Agronomic indicator records:** 77
- **Final merged + cleaned records:** 10,959
- **Training samples:** 7,662 | Test samples: 3,285

2. Data Cleaning Performed

- Removed database IDs and battery metadata columns
- Replaced NULL strings with NaN and imputed with column median
- Rounded timestamps to 10-minute bins; resolved per-line duplicates
- Applied IQR-based outlier detection (factor=3); outliers retained

3. Feature Engineering

- **Total features:** 24
- Features: soil_ec, soil_moisture, soil_temp, co2, air_humidity, air_pressure, air_temp, gdd, standard_day_degree, ontario_units, hour, dow, month, is_daytime, day_sin, day_cos, vpd, roll_sm_1h, roll_ec_1h, roll_co2_1h, temp_diff, sm_change, ec_change, cum_daily_irr
- Temporal: hour, day-of-week, month, is_daytime, cyclic (sin/cos)
- VPD (kPa) from air temperature and humidity
- 1-hour rolling mean for soil moisture, EC, CO₂
- Soil-air temperature differential, soil moisture change, EC change
- Cumulative daily irrigation (lagged by 1 step to avoid leakage)

4. Best Hyperparameters (Optuna)

- **Random Forest:** {'n_estimators': 465, 'max_depth': 5, 'min_samples_split': 20, 'max_features': 'log2'} (CV RMSE=29.374104)
- **XGBoost:** {'n_estimators': 485, 'learning_rate': 0.002657959263519682, 'max_depth': 4, 'subsample': 0.8954425730455775, 'colsample_bytree': 0.8049557256770327} (CV RMSE=24.134206)
- **SVR:** {'C': 98.41675497906036, 'epsilon': 0.0014791631222751686, 'kernel': 'rbf'} (CV RMSE=20.406814)
- **ANN:** {'lr': 0.0015930522616241021, 'u1': 256, 'u2': 32, 'drop': 0.15502135295603015} (CV RMSE=15.892439)

5. Model Performance

	Rank	Model	Type	MAE	RMSE	R ²	MAPE (%)	Train Time (s)	
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	1	CatBoost	ML-Ensemble		2.149974		8.767175	0.7904	54.21 0.56

2	Extra Trees	ML-Ensemble	2.572263	8.775524	0.7900	53.72	0.40	
3	XGBoost	ML-Ensemble	3.239049	9.032593	0.7775	54.47	0.56	
4	Gradient Boosting	ML-Ensemble	2.423044	9.210676	0.7686	58.20	6.44	
5	LightGBM	ML-Ensemble	2.670607	9.293075	0.7645	54.92	2.69	
6	Random Forest	ML-Ensemble	2.194972	9.341222	0.7620	55.49	1.41	
7	RNN	DL	3.144369	10.912043	0.6752	54.51	8.41	
8	RF (Optuna)	ML-Tuned	3.464524	11.561286	0.6354	61.12	0.50	
9	SVR (Optuna)	ML-Tuned	5.452349	12.225621	0.5923	49.63	11.12	
10	XGB (Optuna)	ML-Tuned	4.784351	12.346066	0.5843	62.65	0.56	
11	SVR	ML-Other	3.772672	12.735441	0.5576	54.44	2.14	
12	KNN	ML-Other	3.355026	13.555803	0.4988	65.08	0.00	
13	BiLSTM	DL	4.460432	15.514698	0.3435	77.17	16.44	
14	LSTM (Optuna)	DL-Tuned	4.224822	15.625879	0.3341	78.66	7.95	
15	ANN	DL	7.500152	15.948327	0.3063	53.19	4.51	
16	CNN-1D	DL	8.511388	16.648128	0.2441	72.46	2.38	
17	ANN (Optuna)	DL-Tuned	4.281278	16.914061	0.2197	81.89	2.33	
18	Decision Tree	ML-Ensemble	2.637958	18.252917	0.0913	136.42	0.04	
19	CNN-LSTM	DL	6.871327	18.572542	0.0592	86.81	3.57	
20	LSTM	DL	7.939847	18.750116	0.0411	81.26	8.33	
21	Lasso Regression	ML-Baseline	15.530089	21.198099	-0.2256	70.67	0.36	
22	Linear Regression	ML-Baseline	15.533903	21.203614	-0.2262	70.67	0.01	
23	Ridge Regression	ML-Baseline	15.586363	21.244946	-0.2310	70.65	0.00	

6. Water-Use Efficiency Insights

- The best model (CatBoost) achieves RMSE=8.767175 m³ on a 10-minute window.
- Irrigation volume zero-percentage in test set: 97.0% — models must correctly predict near-zero during rest periods.
- Soil moisture and cumulative daily irrigation are the strongest predictors (see SHAP).
- Rolling features (1-hour) improve performance by capturing sensor dynamics.

7. SDG Alignment

- ****SDG 2 (Zero Hunger):**** Precise irrigation timing → reduced crop stress, higher yield.
- ****SDG 12 (Responsible Consumption):**** ML-optimised scheduling can reduce water usage by 30-50% vs. fixed-timer schedules (FAO 2020), minimising agricultural water footprint.

8. Research Question Findings

- Multi-sensor features consistently outperform single-sensor approaches in preliminary feature importance analysis (SHAP + MI scores show diverse sensor contributions).
- Best ML model: CatBoost | Best DL model: RNN | Best overall: CatBoost
- Tree-based ensembles with Optuna tuning typically lead due to their ability to capture nonlinear threshold effects in irrigation scheduling.
- Deep learning models (LSTM, BiLSTM) capture temporal dependencies in soil sensor data and are competitive, particularly for longer-horizon prediction tasks.

9. Artefacts

- Models: /Users/islambekdaiyn/Desktop/Research/Smart Irrigation/outputs/models
- Figures: /Users/islambekdaiyn/Desktop/Research/Smart Irrigation/outputs/images
- Metrics CSV: /Users/islambekdaiyn/Desktop/Research/Smart Irrigation/outputs/metrics/model_comparison.csv
- This report: /Users/islambekdaiyn/Desktop/Research/Smart Irrigation/outputs/reports/final_report.md